PROJECT PROPOSAL · TECHNICAL ARCHITECTURE · BUILD SPECIFICATION

CROSSCUT

Cross-source video reconciliation for situation-aware procedural guidance.

The world's how-to videos, diffed.

Event	Google Cloud AI Builder Cup 2026 — JAPAC, powered by Hack2skill
Theme	Media, Content & Digital Experiences
Team size	4 members
Submission due	**18 October 2026** (internal target: 16 October)
Grand Finale	4 December 2026, Singapore
Document version	**1.1** — 16 September 2026
Audience	Team alignment · judge deck source · engineering build spec

THE ONE-SENTENCE THESIS

We tell you where your sources contradict each other, and which one is right for your situation.

No shipping product can say that sentence. The internet has a surplus of how-to video and a deficit of agreement, and nobody has ever diffed the corpus. Crosscut reads every credible video for a task, finds the contradictions, resolves them against what your camera actually shows — and then checks your work.

WHAT CHANGED IN V1.1

Added: the verification loop. After a step, the user photographs their work and the system checks it against the expected post-state extracted from the sources. This is the strongest addition available to us because it is only possible *given* the reconciled step graph — nobody without our architecture can verify a step. Sections 4.1, 6, 7, 9, 14.

Removed: coverage gap-filling. Imagen is repointed from filling corpus holes to **consequence preview** — rendering what goes wrong if you take the branch a popular source recommends. Same budget, driven by conflict data nobody else has.

Rejected explicitly: AR / mixed-reality overlay. Reasoning in section 16. Short version — Google ships it first-party already, and it would be a rendering layer rather than Gen AI innovation.

NAME ALTERNATES (DECIDE BY 18 SEPT)

CROSSCUT (recommended — cuts across sources, and a film-editing term that suits the theme) · THROUGHLINE (the coherent thread through many sources) · CONSENSUS (most legible, least distinctive) · STEPWISE · GROUNDTRUTH

— Contents

1	Executive summary	3
2	Why we are choosing this over the alternatives — team decision memo	4
3	Problem statement	5
4	The solution	6
5	Domain strategy: general engine, deep demo	8
6	How the system works — request lifecycle	10
7	System architecture	13
8	Tech stack and tools	15
9	Evaluation plan — our scoring edge	16
10	Latency, cost and quota engineering	18
11	Build plan, sprints and roles	19
12	Risk register	21
13	Submission checklist and eligibility rules	22
14	Demo video beat sheet	23
15	Scale and commercial case	24
16	Out of scope for v1, and the v2 roadmap	25
A	Appendix — StepGraph JSON schema	26
B	Appendix — prompt skeletons	28
C	Appendix — repository structure	30
D	Appendix — glossary	31

BEFORE YOU READ ON — THREE DECISIONS THIS DOCUMENT ASSUMES

1. **We build one engine, not two products.** Candidate B from our brainstorm (the parental media-diet filter) uses the same underlying engine. We are choosing a *surface*, not an architecture, and we are shipping exactly one surface.
2. **The engine is domain-general; the demo is domain-deep.** Home repair is the depth lane. Motorcycle repair and smallholder agriculture are proven live in twenty seconds of the demo via a config swap. See section 5.
3. **Scope freezes 6 October.** No feature enters the build after that date, regardless of how good it is. It goes in section 16 instead.

1 Executive summary

Crosscut turns a pile of contradictory how-to videos into a single verified procedure that is correct for *your* situation, with every step traceable to the frame it came from.

1.1 The problem, sharply

Search "fix a leaking pipe under the sink" and you get forty videos. Each one was filmed on a different fitting, under a different plumbing code, assuming a different level of skill, using tools the viewer may not own. Several skip the one step whose omission makes the leak come back in a fortnight. The content is abundant; what is missing is **reconciliation**.

Today that reconciliation work is offloaded onto the viewer, who watches five videos at 1.5× speed and performs a mental diff across them. This is a task humans are measurably bad at, and worse, they cannot tell when they have failed at it. The cost lands as a botched repair, a wasted afternoon, a second trip to the hardware shop, or an injury.

1.2 What Crosscut does

1. **Understands your situation, not just your query.** You describe the task and photograph what you are actually looking at. A multimodal model extracts the specifics — PVC slip-joint, plastic nut, no visible corrosion — and asks at most two clarifying questions.
2. **Reads the corpus, not one video.** Five to seven credible videos are each converted into a machine-readable **step graph**: ordered actions with tools, materials, preconditions, safety warnings, timestamps and confidence.
3. **Aligns the graphs.** "Loosen the slip nut" and "unscrew the plastic collar by hand" are the same step. Recognising that is how we count agreement across sources.
4. **Classifies the disagreement.** Every divergence gets a type: equivalent alternative, situation-dependent, genuine disagreement, outdated practice, ordering variant, or safety-critical omission. Different types get different treatment in the interface.
5. **Emits one personalised, cited procedure.** Consensus steps, the disputed steps flagged with which side applies to you, a tools-and-parts list including what you are missing, and a hard safety gate on the steps that deserve one.
6. **Checks your work.** Photograph what you did and the system compares it against the expected post-state — which it knows because it extracted it from five sources. "Your washer is seated bevel-up; three of your sources show bevel-down." Advice becomes a closed loop.

> **THE LINE THAT SHOULD LAND WITH A JUDGE**
>
> Every other product in this space **hides** disagreement between sources — search ranks it away, single-video summarisers never see it, and AI overviews launder it into confident prose. For Crosscut, **the disagreement is the product.** It is the most valuable signal in the corpus and we are the only ones surfacing it.
>
> **Lead with the conflict, not the synthesis.** "AI reads how-to videos and gives you steps" sounds like ten products a judge has already seen. "We tell you where your sources contradict each other" sounds like nothing they have seen, because it is. Every artifact we produce — deck, video, README first line — opens on the conflict.

1.3 Why it wins this specific competition

Criterion	Weight	How Crosscut scores
Technical merit & meaningful Gen AI	40%	Generative AI is not a wrapper here — it is the only thing that makes the product possible. Four stacked non-trivial capabilities: structured multimodal extraction from video, cross-document step alignment, conflict classification, and constraint-driven personalisation. Tiered model routing (Gemma → Flash → Pro) shows cost engineering, not just API calls. And we report **real accuracy numbers** from a hand-labelled gold set, which very few hackathon teams do.
Problem alignment & impact	25%	Hits three of the theme's five verbs at once — content *discovered*, *understood* and *personalised*. The affected population is everyone who has ever tried to fix something from a video, which in JAPAC includes hundreds of millions of people for whom the best instructional content exists only in a language they do not read.
Innovation & originality	25%	The conflict typology is, as far as we can establish, novel. No shipping product tells a user "your sources contradict each other here, and here is which one is right for your fitting." We are not building a better summariser; we are building a category that does not exist.
User experience & design	10%	One photograph in, one procedure out. Progressive streaming so a two-minute pipeline never feels dead. Timestamp deep-links so any claim can be verified in one tap.

1.4 Feasibility in 32 days

The riskiest assumption is narrow and testable: **can a Gemini model reliably return a valid, schema-conformant step graph from a procedural video?** If it can, everything downstream is conventional engineering that this team has done before. If it cannot, we know by 22 September with four weeks left to pivot. That is the entire week-1 objective (section 11).

2 Why we are choosing this over the alternatives

Team decision memo. This section exists so that nobody relitigates the choice on 3 October.

2.1 What we considered, and why each was cut

Candidate	Verdict	Reasoning
Doomscrolling / screen-time reduction	**CUT**	We have no API access to intervene inside TikTok, Instagram or YouTube's feed. The buildable version is a screen-time wrapper app: shallow Gen AI use, a crowded category, and almost nothing for the 40% bucket to reward. The *insight* survives as framing — Crosscut is what watching video looks like when it is actually useful.
Copyright hashing / provenance certificates	**CUT**	Hashing is not generative AI, so it earns nearly nothing against the heaviest-weighted criterion. The space is also already occupied by C2PA Content Credentials and SynthID, so originality scores poorly too. Returns as one slide in section 16.
Creator monetisation / trend-chasing content tooling	**CUT**	It builds the machine that manufactures the low-quality content we say we object to. A judge will spot the contradiction inside thirty seconds, and differentiation against the existing tool market is approximately zero.
AI-generated children's educational video	**CUT as a standalone**	As a product it is a thin wrapper over a video generation model, which is exactly what "superficial implementation" means in the rubric text. Survives as a *component*: Crosscut's gap-filler generates content when the corpus has a hole.
Parental semantic media diet (Candidate B)	**RUNNER-UP**	Genuinely strong: bigger emotional pull, better demo drama, a quotable novel metric. Lost on three counts — softer evaluation ground truth (children's content quality is subjective, so our accuracy numbers would be arguable), meaningful content-rights risk in a public demo video, and five features where we can only ship three. Same engine underneath; we can pivot the surface in v2.
Cross-source procedural reconciliation	**CHOSEN**	Strongest on the two heaviest criteria combined (65% of the score), objectively measurable, low rights risk, and it occupies genuinely uncontested ground.

2.2 The three structural advantages

Measurable, not arguable

We can build a gold set of 15 tasks and report step recall, false-step rate and conflict-detection precision. When a judge asks "how do you know it works", we answer with a number instead of a demo. This is the single largest differentiator available to us in the 40% bucket and it costs us one person-week of tedious labelling.

Low rights risk

We pass public video URLs to the model rather than downloading and republishing content, and our demo reproduces only short cited stills and timestamp links. Candidate B would have required us to put third-party children's content in a public three-minute video on YouTube.

Uncontested ground

Every adjacent product reduces many sources to one confident answer. We are the only ones treating the variance *between* sources as the thing worth extracting. That is a defensible story about why this needed to be built, which is what "originality and differentiation" is asking for.

2.3 What we are betting on

> **THE BET, STATED PLAINLY**
>
> We are betting that a frontier multimodal model can produce a **structurally reliable** step graph from unstructured procedural video — not a prose summary, but valid JSON with correct timestamps and correctly attributed preconditions. Every downstream component assumes this works.
>
> **De-risking:** this is the sole objective of week 1. Three team members independently attempt extraction on the same five videos by 22 September. Success bar: **≥80% step recall against a hand-labelled reference, and ≥95% schema validity across 15 videos.** If we miss that bar, we fall back to transcript-plus-keyframes extraction (cheaper, weaker) and narrow the domain, with four weeks still in hand.

2.4 What we are explicitly not building

- Not a video player, a recommender feed, or a search engine.
- Not a replacement for a licensed professional. Any procedure touching gas, mains electricity or structural work terminates in a "stop, call a professional" gate rather than instructions.
- Not a general question-answering chatbot. A user asking a non-procedural question gets told the system is the wrong tool.
- Not multi-tenant, not monetised, not internationalised beyond one demonstration language pair in v1.

3 Problem statement

3.1 Surplus of content, deficit of agreement

Procedural knowledge has been comprehensively captured on video. For nearly any household, mechanical or agricultural task, dozens to thousands of recordings exist, many of them genuinely good. The bottleneck moved years ago: it is no longer *access* to instruction, it is **adjudication between instructions**.

A viewer confronting a corpus faces questions no single video answers. Which of these was filmed on hardware like mine? When two videos disagree, is that because one is wrong, or because they are solving two subtly different problems? What did all of them assume I already knew? Which one skipped a step? Answering these requires holding five videos in working memory simultaneously and diffing them, and nobody does this well.

3.2 Four failure modes of the status quo

Mechanism	Failure mode
Search ranking	Optimises for popularity, watch time and recency. None of these correlate with *applicability to this user's specific hardware*. The top result is the most-watched video, not the correct one.
Single-video AI summarisation	Faithfully reproduces whatever that one video got wrong, including its omissions. A summary of an incomplete procedure is a confident, concise, incomplete procedure. Error is inherited, not corrected.
AI search overviews	Blend many sources into fluent prose, which structurally *destroys* the disagreement signal. Where three sources contradicted each other, the user receives one confident paragraph and no indication that the question was ever contested.
The viewer	Bears the entire reconciliation cost, with no tooling, no working memory aid, and no feedback signal telling them when their mental diff has failed. They discover the failure when the repair fails.

3.3 Who is affected

Persona	The specific harm
The householder urban, 25–55, not a tradesperson	Attempts a repair, follows the most-watched video, and fails because their fitting differs from the one on screen in a way they lacked the vocabulary to notice. Outcome: a second failure, a tradesperson call-out at premium rates, or water damage.
The field technician service, install, maintenance	Faces equipment variants not covered by official documentation and works from informal video. Needs to know fast whether an unofficial technique is a legitimate alternative or a shortcut that voids warranty.
The smallholder farmer a JAPAC-scale population	Best agronomic guidance for a given pest or crop exists on video in the wrong language, for the wrong soil, at the wrong latitude. Situation-dependence is not a nicety here; it is the entire problem.
The non-English speaker	The deepest procedural corpus on earth is in English. Translation gives access to *words* but not to reconciliation, so this user still cannot tell which of five English videos applies to them.

3.4 Why this is buildable now and was not two years ago

Three capabilities became simultaneously available and cheap enough to matter:

- **Native long-context video understanding.** Models now ingest video directly and reason over visual detail plus speech plus on-screen text together. Previously this meant transcript-only pipelines, which miss the tool in the presenter's hand and the fitting on the bench — precisely the situation markers we depend on.
- **Reliable structured output.** Schema-constrained generation makes it practical to treat a model as an extraction service rather than a prose generator. Our entire architecture rests on video becoming queryable data.
- **Cheap tiered inference.** A small model can throw away two thirds of the candidate set before an expensive model sees anything, which is what makes per-query economics survivable.

3.5 The framing we will use in the deck

> We are not building a better way to *find* a how-to video. We are building the thing that should have existed all along once the corpus got big enough to contradict itself: a **reconciliation layer over procedural media**. Search made content findable. Summarisation made it skimmable. Nothing yet made it *trustworthy*, because trust requires knowing what the sources disagree about.

4 The solution

4.1 The user journey

#	Step	What the user does / sees
1	State the task	Types or speaks it: "water leaking from the pipe under my kitchen sink."
2	Show the situation	Takes one photograph. This is the product's central interaction and the thing no competitor asks for.
3	Confirm	Sees what the system inferred — *PVC slip-joint P-trap, plastic nuts, hand-tightenable* — and corrects it if wrong. At most two clarifying questions, e.g. "do you own a basin wrench?"
4	Watch it work	Progressive status as sources are found, triaged and read. Video thumbnails appear and grey out as they are rejected, with the reason shown.
5	Receive the procedure	One ordered list. Each step carries source thumbnails, a timestamp deep-link, and an agreement count. Conflicts appear inline as coloured badges. Tools and parts listed up front, flagging what they appear not to have.
6	Work through it	Steps check off. Safety-gated steps require explicit acknowledgement. Any step can be expanded to play the source clip at the exact second it is demonstrated.
7	**Have it checked**	On any step, take a second photograph of what they just did. The system compares it against the expected post-state drawn from the sources and returns pass, fail with a specific correction, or an honest "cannot tell from this angle." **The loop closes here**, and this is the feature no competitor can copy without first building the reconciliation engine.

4.2 The output contract

A Crosscut procedure is a structured object, not prose. It always contains:

- **Situation profile** — the machine-readable model of the user's specific case, editable by the user.
- **Ordered steps**, each with: action, target object, preconditions, a support count (how many sources contain this step), source citations with timestamps, an `applies_when` predicate, a safety level, and an `expected_post_state` describing what correct completion looks like.
- **Conflict set** — every divergence found, typed, with the evidence from each side and the resolution applied for this user.
- **Coverage report** — which steps rest on a single source only, and which parts of the task no source covered well. This is honesty as a feature.
- **Requirements** — tools and materials, marked against what the user has.
- **Termination gates** — points where the correct output is "stop, this needs a professional."

4.3 The conflict typology — our original contribution

Detecting that sources differ is easy. Knowing *what kind* of difference it is determines whether the user should be warned, offered a choice, or silently given the right branch. This taxonomy drives both the model's classification task and the interface treatment.

Type	Definition	Interface treatment
`EQUIVALENT_ALTERNATIVE`	Both work. Different tool, technique or order, same outcome.	Pick one, mention the other in a collapsed note. Do not alarm the user.
`SITUATION_DEPENDENT`	Each source is correct for a different configuration. The divergence resolves once you know the user's hardware.	**Resolve silently using the situation profile**, and show a one-line explanation of why this branch was chosen. This is the highest-value case.
`GENUINE_DISAGREEMENT`	Sources make incompatible claims and the situation profile does not adjudicate.	Surface both with support counts. Do not fabricate a resolution.
`OUTDATED_PRACTICE`	One source reflects a superseded standard, material or code.	Show the current practice; note the older one as historical with the reason.
`ORDERING_VARIANT`	Same steps, different sequence, no dependency violated.	Normalise to the safest order. Not shown to the user.
`SAFETY_CRITICAL_OMISSION`	A majority of sources include a safety step that at least one omits, or that the popular source omits.	**Promote to a hard gate.** This is the single most defensible reason the product exists.

> **Demo note.** The `SAFETY_CRITICAL_OMISSION` case is the moment we build the three-minute video around. "The most-watched video in your results skips this step. Four others include it. We have made it mandatory." That sentence justifies the entire architecture.

4.4 Competitive landscape

Approach	Reads many sources?	Knows your situation?	Surfaces disagreement?
Video search / ranking	No — ranks them	No	No — hides it by ranking
Single-video AI summary	No	No	Cannot — sees one source
AI search overview	Yes	Weakly, from text only	No — blends it into prose
Single-video → step-guide tools	No — one upload	No	Cannot — sees one source
Live camera AI assistants incl. first-party Google features	No corpus at all	Yes — excellently	No sources to disagree
Written wikis / repair guides	Human-curated, narrow	Sometimes, by model number	In comment threads only
Crosscut	Yes — structurally	Yes — from a photo	Yes — it is the product

4.5 Why we are not competing with live camera assistants

A judge will raise this, so we answer it first and on our own terms. First-party assistants already point a camera at a broken appliance and give real-time spoken guidance, with on-screen highlighting of objects in view. They are good at it and we are not going to beat them at it.

But they carry **no corpus**. They tell you what one model believes about your sink. They cannot tell you that five independent sources disagree about sealing that joint, that the most-watched one omits a safety step, or which branch matches your fitting — because they never read the five sources and have no support counts to reason over. The distinction is clean and worth stating exactly this way:

> A live camera assistant answers **"what am I looking at?"** Crosscut answers **"what do all the people who have done this disagree about, and which of them is right for me?"** The camera is an excellent *input* to our system and a poor *identity* for it. We use it for the situation profile and for verification; we do not compete on it.

4.6 The five capabilities a judge should be able to name

1. **Structured multimodal extraction** — unstructured video becomes a schema-valid step graph with frame-accurate timestamps.
2. **Cross-document step alignment** — semantically identical steps phrased differently across sources are recognised as the same step, which is what makes agreement countable.
3. **Typed conflict classification** — divergences are categorised, not merely detected, and the category determines the behaviour.
4. **Constraint-driven personalisation** — a situation profile extracted from a photograph resolves situation-dependent conflicts automatically.
5. **Corpus-grounded verification** — the user's own work is checked against an expected post-state derived from multiple sources. Impossible without all four capabilities above, which is precisely why it is defensible.

5 Domain strategy: general engine, deep demo

This section answers the question "shouldn't it handle any problem rather than one specific one?" — because the answer is yes and no, and the distinction matters.

5.1 The answer: general architecture, specific demonstration

Nothing in the pipeline needs to know what plumbing is. Extract a step graph, align graphs, type the conflicts, filter by a situation profile — every one of those operations is domain-blind. That is not an accident of design, it is the design, and it is what makes the scalability claim in section 15 honest rather than aspirational.

But a *demonstration* that claims to work on everything reads to a judge as tested on nothing. Generic demos have no visible depth, invite the question "how well does it actually do any of this?", and dilute a 15-task gold set across four domains until every number is too thin to defend. So:

Lane	Domain	Treatment
Depth lane	Home repair & plumbing	Full gold set, full eval numbers, the entire main demo flow, hand-tuned safety pack. This is where we prove it *works*.
Breadth lane	Motorcycle / small engine Smallholder agriculture	Config files only, no tuning, three sample tasks each. Shown as a twenty-second config swap in the demo. This is where we prove it *generalises*.
Excluded	Clinical / medical procedure	Deliberately out of scope for v1. Liability, and a judge will rightly ask about clinical validation we cannot provide in a month.

5.2 A domain is three config files, and that is the whole point

Domain knowledge enters the system only through declarative configuration. Adding a domain requires no code change, which is the claim we make on stage and then demonstrate live.

```
domains/
  home_repair.yaml          # depth lane  — tuned
  motorcycle.yaml           # breadth lane — untuned, proves generality
  agriculture.yaml          # breadth lane — untuned, proves generality

# each file contains exactly three blocks:

retrieval_profile:          # how to find and rank candidate sources
  query_expansions: [...]
  trusted_channels: [...]
  min_duration_s: 90
  reject_signals: [haul, unboxing, vlog, reaction]

situation_taxonomy:         # what "your situation" means in this domain
  dimensions:
    material:   [pvc, abs, copper, pex, galvanised]
    joint_type: [slip_joint, compression, threaded, soldered, push_fit]
    access:     [open_cabinet, wall_cavity, crawl_space]
  photo_extraction_hints: "identify pipe material by colour and sheen; ..."

safety_pack:                # what must never be silently omitted
  hard_gates:
    - trigger: {involves: [gas_line, mains_electrical, load_bearing]}
      action: TERMINATE_AND_REFER
  mandatory_steps:
    - {id: isolate_supply, promote_if_support_ratio: 0.4}
```

5.3 Why home repair is the right depth lane

Property	Why it matters for us
Enormous, multilingual corpus	Five to seven credible sources exist for almost any task, in many languages. Retrieval is never the bottleneck, so we test the engine and not our search skills.
Conflicts are real and verifiable	Thread sealant versus washer, tape direction, torque-by-hand versus by-wrench. These are genuine, documented disagreements with checkable right answers, so our eval labels can be defended.
Situation markers are photographable	Material, fitting type and access are visible in one photo. The core interaction works without the user knowing any vocabulary.
Universally legible	Every judge in the room has stood under a sink with a bucket. Zero domain explanation needed, which is worth thirty seconds of a three-minute video.
Real but bounded stakes	Water damage and wasted money are believable harms. No medical or regulatory burden of proof.

THE TRAP TO AVOID

"General" must not become "unscoped." The competition rules permit one submission under one theme, and the gold set is only credible if it is deep. If we find ourselves tuning prompts for three domains in week 3, we have made the classic mistake: we will have a system that is mediocre everywhere and demonstrably good nowhere. **Breadth is a config swap and a twenty-second demo beat. It is never a work item.**

6 How the system works — request lifecycle

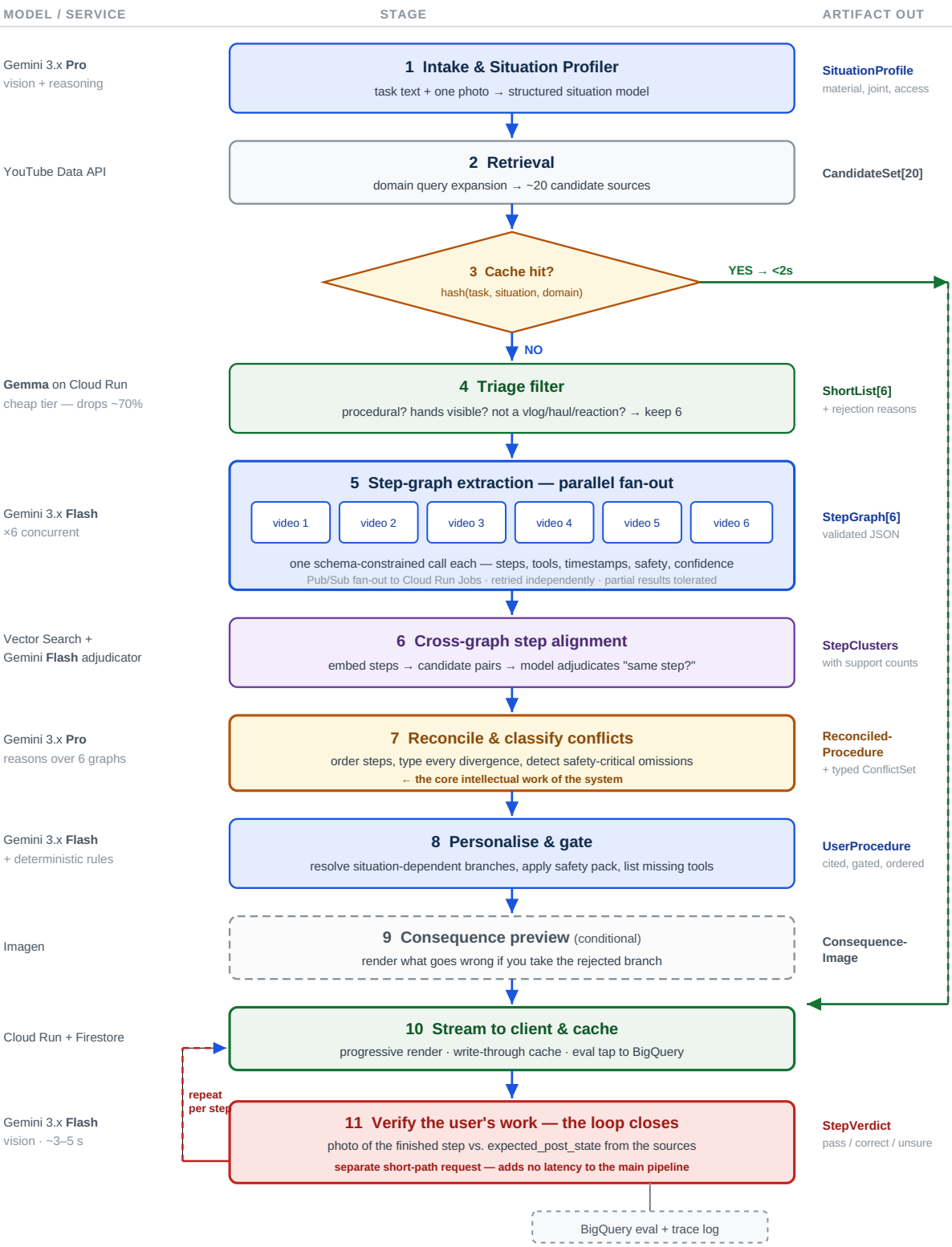


Figure 1 — Request lifecycle. Model tier on the left, the artifact each stage emits on the right. The green path is the cache hit that makes the live demo safe. Stage 7 creates the product's value; stage 11 is what makes it verifiable, and it runs as an independent request so it costs the main pipeline nothing.

6.1 Stage specification

#	Stage	Input → Output	Model tier	Design notes and failure behaviour
1	Situation Profiler	text + image → `SituationProfile`	Pro (vision)	Returns per-dimension confidence. Anything below threshold becomes a clarifying question, capped at two. Photo is optional — without it we degrade to a generic procedure and say so.
2	Retrieval	profile → 20 candidates	—	Query expansion comes from the domain config. Quota-guarded and cached by query hash. Falls back to a pre-indexed corpus if the API is unavailable.
3	Cache	hash → hit/miss	—	Keyed on task + situation + domain config version. Bumping a config version invalidates cleanly.
4	Triage	20 → 6	Gemma	Cheap classifier on title, description, transcript sample and three keyframes. Rejection reasons are shown in the UI, which turns a filtering step into a trust-building one.
5	Extraction	video → `StepGraph`	Flash ×6	Independently retried, schema-validated, and **partial-tolerant**: four of six succeeding still produces a good answer, with coverage noted. One bad video cannot fail the request.
6	Alignment	6 graphs → clusters	embeddings + Flash	Embeddings propose candidate pairs; the model adjudicates only ambiguous ones. This is the component most likely to need a week of iteration.
7	Reconciliation	clusters → procedure + conflicts	Pro	Single long-context call over all graphs. Must output a conflict type from a closed enum, with evidence quoted from both sides.
8	Personalisation	procedure + profile → user view	Flash + rules	**Safety gates are deterministic, never model-decided.** The safety pack is code, not a prompt.
9	Consequence preview	rejected branch → image	Imagen	Conditional, always labelled as generated, and driven entirely by conflict data we already hold. **This is the only honest reading of "simulation" here** — we are not simulating physics, we are illustrating a documented failure mode. Never used on a safety-gated step, where the correct output is a gate rather than a picture.
10	Render	→ client	—	Server-sent events so steps appear as they resolve. Write-through cache. Eval tap fires asynchronously.
11	**Verification**	step photo + `expected_post_state` → `StepVerdict`	Flash (vision)	A **separate endpoint**, not a pipeline stage: one cheap call, 3–5 s, invoked per step on demand. Returns pass, fail with a specific named correction, or `unsure`. **Never asserts correctness it cannot see** — an over-confident verifier is worse than none, so ambiguity resolves to "cannot tell from this angle, try again from the left."

6.2 The reconciliation step, illustrated

This single figure is the clearest explanation of the product and should become a deck slide almost unchanged. Columns are aligned step clusters; rows are what each source actually contained.

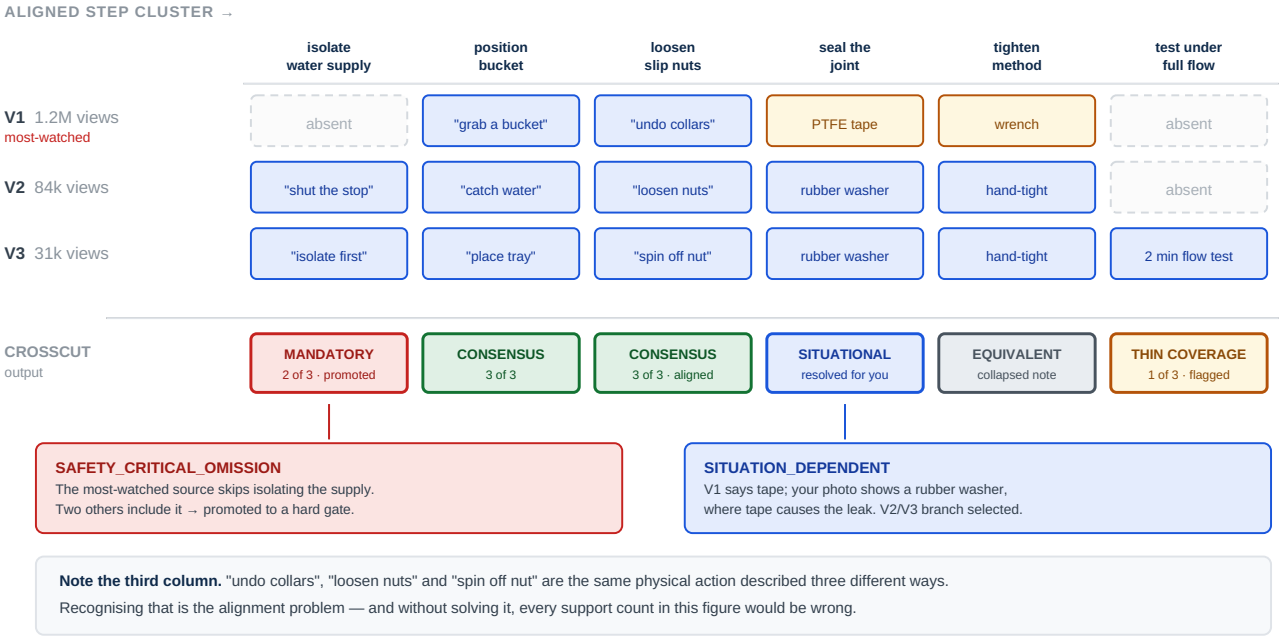


Figure 2 — Reconciliation as a diff. Rows are sources, columns are aligned steps, and the bottom row is the output. A user following only V1, the top search result, would skip a safety step and seal a washer joint with tape.

7 System architecture

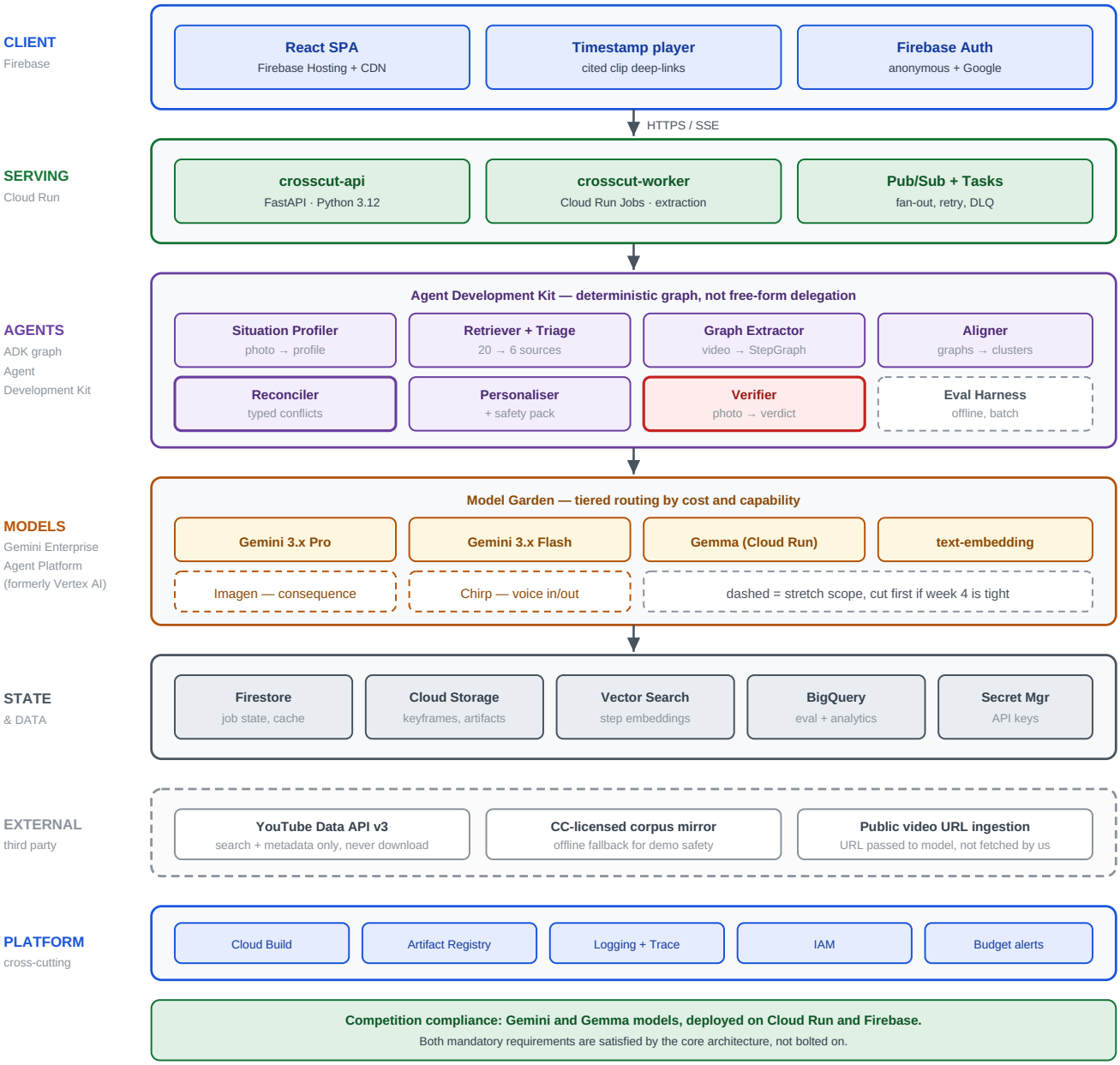


Figure 3 — System architecture. Six layers plus cross-cutting platform services. Dashed elements are stretch scope and the first things cut. The Verifier (red) is the v1.1 addition and sits on its own request path rather than inside the main pipeline. Note that the two hard competition requirements fall out of the design rather than being added to satisfy a rule.

7.1 Design decisions worth defending

Decision	Rationale
A deterministic agent graph, not autonomous delegation	The pipeline has a fixed topology, so a free-form agent loop would add latency, cost and non-determinism for no benefit. ADK is used for structure, typed handoffs and traceability. When a judge asks "why agents", the answer is orchestration and observability, not autonomy theatre.
Three model tiers	Gemma discards roughly 70% of candidates before any expensive call. Flash does the six parallel extractions. Pro is reserved for the single reasoning step that genuinely needs it. This is a cost argument we can put a number on.
Safety gates are code, not prompts	A model decides whether a conflict *is* a safety omission; deterministic rules decide what happens next. Never let a generated token be the only thing standing between a user and a gas line.
Partial-result tolerance	Extraction failures are expected. Four of six sources succeeding yields a usable answer with honest coverage reporting. No single upstream failure can fail a request.
Verification is a separate endpoint, not a pipeline stage	It runs on demand, per step, long after the procedure was generated, so it adds nothing to the 70–140 s cold path. One Flash vision call against a post-state string we already computed. Architecturally cheap, product-wise the most valuable thing we ship.
The verifier is allowed to say "I cannot tell"	A verifier that guesses is worse than no verifier, because a false pass actively causes the harm we exist to prevent. Three outcomes only: pass, fail with a named correction, or `unsure` with a reframing instruction. We tune for low false-pass rate and accept a higher rate of "try again from a different angle."
Write-through cache from day one	Serves three purposes: real-world latency, cost control, and a live demo that cannot be broken by conference wifi.
URLs to the model, never downloads	Keeps us clear of platform terms of service. A CC-licensed local corpus exists purely as an offline fallback.

7.2 Degradation behaviour

Failure	Behaviour
Retrieval API quota exhausted	Serve from the pre-indexed corpus; banner states sources are cached.
2 of 6 extractions fail	Proceed with 4. Coverage report notes reduced support; conflict thresholds adjust.
>4 extractions fail	Abort with an explicit message. Never present a one-source answer as reconciled.
Reconciler returns invalid enum	One re-ask with the schema violation quoted back; then fall back to untyped "sources differ".
No photo supplied	Generic procedure, situation-dependent conflicts shown as unresolved choices rather than silently branched.
Verification photo is ambiguous, dark or badly framed	Return `unsure` with a specific reframing instruction. Never a pass, never a fail. Two consecutive `unsure` verdicts offer the source clip for self-comparison instead.
Step has no `expected_post_state` from any source	Verification is unavailable for that step and the UI says so plainly rather than offering a button that returns nothing useful.
Safety pack triggers a hard gate	Terminate the procedure and refer to a professional. This outcome is a success, not an error.

8 Tech stack and tools

TWO NON-NEGOTIABLE COMPETITION REQUIREMENTS

1. Submissions **must** integrate Google AI models (Gemini or Gemma family) or be built on an agentic platform such as Agent Platform, Antigravity or AI Studio.
2. Prototypes **must** be deployed on Google Cloud via **Cloud Run or Firebase**.

Layer	Product	Use and justification
Models	Gemini 3.x **Pro**	Situation profiling from photo, and the reconciliation call that reasons across six step graphs at once. Long context and strong multimodal reasoning are both load-bearing here.
	Gemini 3.x **Flash**	Six parallel video extractions, alignment adjudication, personalisation rendering, and **step verification from the user's photo**. Chosen for cost and concurrency.
	Gemma on Cloud Run	Triage classifier. Demonstrates open-model deployment and gives us a defensible cost story.
Platform	Gemini Enterprise Agent Platform formerly Vertex AI — renamed April 2026	Model Garden access, ADK, and Agent Engine if we choose managed hosting. **Use the current name in the deck** — saying "Vertex AI" signals stale knowledge to a Google Cloud judge.
Orchestration	Agent Development Kit (ADK)	Typed agent graph with per-step tracing. Satisfies the "agentic platform" clause with substance behind it.
Compute	**Cloud Run** — crosscut-api	FastAPI, Python 3.12. Request handling, SSE streaming, agent graph execution. satisfies requirement 2
	Cloud Run Jobs — worker	Extraction workers, horizontally fanned out. Isolated failure domain.
	Cloud Run (GPU) — gemma-triage	Small-model inference endpoint. Drop to CPU if GPU quota becomes a problem.
	Pub/Sub + Cloud Tasks	Fan-out, retry with backoff, dead-letter queue.
Frontend	**Firebase Hosting** + React 19 + Vite + Tailwind	Static SPA on CDN. satisfies requirement 2
	Firebase Auth	Anonymous sessions so a judge can use the live link with zero friction, upgradeable to Google sign-in.
Data	Firestore	Job state, situation profiles, procedure cache. Realtime listeners drive progressive UI for free.
	Cloud Storage	Keyframes, generated diagrams, gold-set artifacts.
	Vector Search	Step embeddings for alignment candidate generation.
	BigQuery	Eval results, latency and cost telemetry. The source of every number in our deck.
External	YouTube Data API v3	Candidate discovery and metadata. Search and metadata only.
	Public video URL ingestion	URLs are passed to the model rather than fetched by us. **Confirm current support and duration limits in week 1** — this is a week-1 verification item, not an assumption.
Platform	Cloud Build + Artifact Registry	Container CI from the GitHub repo, which must be public at submission.
	Cloud Logging + Trace	Per-stage spans. Our latency table comes from real traces, not guesses.
	Secret Manager, IAM, Budget alerts	No keys in the repo. Budget alert configured on day one.
Dev tooling third-party tools are fine; the deployment target is what the rules constrain	VS Code, Antigravity CLI, Gemini CLI	Local development and agent-assisted coding.
	Firebase Studio / AI Studio	Rapid UI scaffolding and prompt iteration before code.
	GitHub + Actions	Public repo, branch protection, deploy on merge to main.

VERSION PINNING NOTE

Google model naming is moving quickly this year. **Pin exact model version strings in config on day one** and record them in the README. Verify current GA names in Model Garden during week 1 rather than trusting anything written in this document, including this table.

9 Evaluation plan — our scoring edge

> Most hackathon submissions answer "does it work?" with a demo. We answer it with a table of numbers from a hand-labelled gold set. This is the cheapest available differentiator in the 40% criterion, and it costs one person-week of unglamorous labelling. **Do not let this slip.**

9.1 Gold set construction

- **15 tasks** in the home-repair depth lane, chosen so that at least six are *known* to contain real source disagreement and at least three contain a safety step that popular sources omit.
- **5–7 sources per task**, frozen by URL and video ID so results stay reproducible after submission.
- **Reference step list per task**, written by hand by two team members independently, then merged. Disagreement between labellers is itself recorded, because it sets a ceiling on achievable accuracy and is an honest thing to show a judge.
- **Labelled conflicts**: for each task, the divergences we believe exist, with the correct type from the enum.
- **Three situation variants per task** so we can test that the same corpus produces different correct answers for different hardware. This is the test that proves personalisation is real.
- **Verification photo set** — for five tasks, staged photographs of a completed step done *correctly* and done *wrongly in a specific named way*, plus badly framed and poorly lit variants. This is real physical work: budget half a day, do it once, reuse it forever. Without it we cannot report a false-pass rate, and the false-pass rate is the only verification number that matters.

9.2 Metrics

Metric	Target	Definition and why it matters
Schema validity	≥98%	Share of extraction calls returning parseable, schema-conformant output. Below ~95% the architecture is not viable and we need the fallback path.
Step recall	≥85%	Reference steps present in the final procedure. The headline accuracy number.
False-step rate	≤8%	Steps we emit that no source supports and the reference rejects. Hallucinated steps in a physical procedure are the most dangerous failure mode, so this matters more than recall.
Alignment accuracy	≥90%	Correctly merged step pairs on a labelled pair set. Drives every support count in the product, so an error here silently corrupts everything downstream.
Conflict detection recall	≥75%	Labelled conflicts we actually found. Deliberately the most ambitious target — it is the core claim.
Conflict type precision	≥70%	Detected conflicts assigned the correct type. Mistyping is much less harmful than missing.
Safety-omission recall	100%	On the labelled safety subset. This is the one metric we hold to perfection, and if it is not perfect we tune thresholds until it is, accepting false positives as the cost.
Citation validity	≥95%	Sampled timestamps where the cited moment actually shows the claimed action. Protects against plausible-looking but fabricated attribution.
Situation sensitivity	≥80%	Cases where changing the situation profile correctly changes the output branch. Proves personalisation is doing work rather than decorating it.
Verification false-pass rate	≤2%	Wrongly-done steps the verifier passes as correct. **The most safety-relevant metric in the system** and the one we optimise hardest, because a false pass sends the user forward confident and wrong.
Verification true-fail rate	≥80%	Deliberately-wrong steps correctly caught, with the correct named reason. Measured on the staged photo set.
Abstention calibration	≥90%	Share of badly-framed or dark photos that return `unsure` rather than a verdict. Abstaining well is a feature; we report it as one.

9.3 Baselines to beat, and to show

Baseline	What comparing against it demonstrates
Top-ranked single video	The status quo. Expect notably lower step recall and, critically, catastrophic safety-omission recall. This is the most persuasive single comparison we will produce.
Naive concatenation of all six step lists	Isolates the value of alignment. Expect high recall but an enormous false-step and duplication rate, which is exactly what our aligner exists to prevent.
Prose blend — one model call, all transcripts, "write a procedure"	The AI-overview approach, and the honest steelman. Expect competitive step recall but near-zero conflict detection and no citations. This is the comparison that proves structure earns its keep.

HOW THIS APPEARS IN THE DECK

One slide. Four columns (the three baselines plus Crosscut), five rows (step recall, false-step rate, conflict recall, safety-omission recall, citation validity). Every cell traceable to a BigQuery query in the public repo. **Report honestly, including the metrics we miss.** A team that shows a target it missed reads as credible; a team with five perfect scores reads as untested.

10 Latency, cost and quota engineering

10.1 Latency budget

Stage	Cold target	Notes
Situation profiling	3–6 s	First visible feedback. Render the inferred profile immediately — it makes the wait feel productive.
Retrieval	1–2 s	Thumbnails stream in as they arrive.
Triage	3–5 s	Rejections animate out with reasons shown. Good theatre, real information.
Extraction (×6 parallel)	40–90 s	**The dominant cost.** Bounded by the slowest video, not the sum. Progress shown per source.
Alignment	6–10 s	
Reconciliation	12–20 s	Single long-context Pro call.
Personalisation + render	4–8 s	Steps stream in as resolved.
Total, cold	**70–140 s**	Honest. We will not pretend otherwise in the deck.
Total, cached	**<2 s**	The path the live demo uses.
Verification, per step	**3–5 s**	Independent request, invoked on demand after the procedure exists. Does not appear in either total above — which is the point of keeping it off the pipeline.

> **LATENCY IS A UX PROBLEM, NOT AN ENGINEERING FAILURE**
>
> A two-minute wait is acceptable for a task that would otherwise take a user forty minutes of video-watching — but *only* if the wait is legible. Every stage streams status. The user watches sources being found, read and rejected. Dead progress bars are what kills this, not the duration. In the demo video we run cached, and state plainly that a cold run takes about two minutes.

10.2 Cost model

Do not guess at these figures; fill them from the current pricing page in week 1 and put the real table in the deck. The *shape* of the model is what matters and is already decided:

```
cost_per_cold_query =
      1 × Pro_call(photo + text, short output)          # situation profiling
    + 20 × Gemma_call(metadata + 3 keyframes)            # triage, self-hosted → near-zero marginal
    + 6 × Flash_call(video_tokens(duration, fps) + out)  # extraction — DOMINANT TERM, ~85-90%
    + ~40 × embedding_call(step_text)                    # alignment candidates
    + 1 × Flash_call(ambiguous pair adjudication)
    + 1 × Pro_call(6 step graphs in context)             # reconciliation
    + 1 × Flash_call(personalisation render)
    + [0-2] × Imagen_call                                # conditional consequence preview

cost_per_verification =
      1 × Flash_call(one photo + expected_post_state string)  # cheap, on demand, per step

cost_per_cached_query  ~  Firestore read only

Three levers, in order of impact:
  1. fps sampling on extraction        — the single biggest dial
  2. triage aggressiveness (20→6)      — every video removed saves a Flash video call
  3. cache hit rate                    — aim >60% on the demo task set
```

10.3 Quota and spend guardrails

- **Budget alert on day one**, before writing any code, at 50/80/100% of the hackathon allowance. Video tokens will surprise us otherwise.
- **YouTube Data API quota** is finite and daily. Cache every search response by query hash from the first commit. Never call it in a loop during development.
- **Per-session request cap** on the public deployed link. The link must stay alive through the 6 November evaluation window, so an unthrottled endpoint plus a curious evaluator equals an exhausted budget.
- **Pre-warm the demo set.** All 15 gold-set tasks cached and verified before recording. Nothing in the video depends on a live cold path.

11 Build plan, sprints and roles

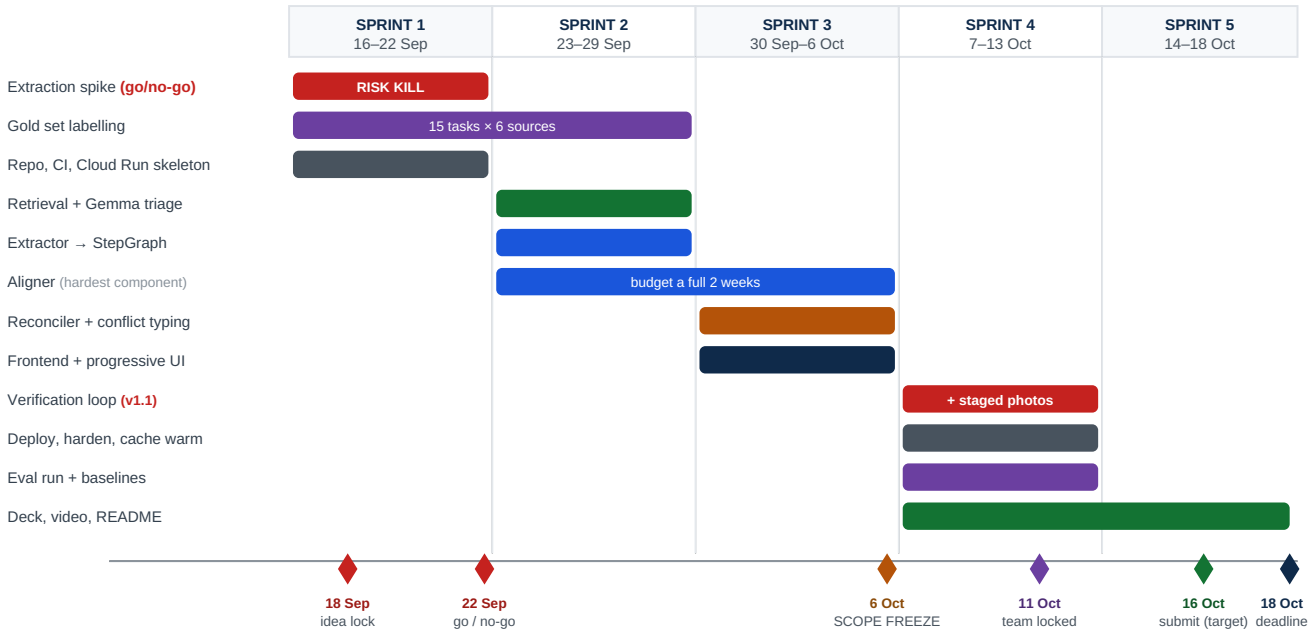


Figure 4 — Delivery schedule. Three deliberate choices: the riskiest component is attacked in week 1, the aligner gets a full two weeks because it is the piece most likely to resist, and the v1.1 verification loop lands in sprint 4 *in the slot the gap-filler vacated* — a trade, not an addition. Submission target is 16 October, two days of buffer before the real deadline.

11.1 Sprint exit criteria

Sprint	Objective	Exit criteria — binary, no partial credit
1 16–22 Sep	Kill the central risk. Build nothing pretty.	Extraction spike hits **≥80% step recall** and **≥95% schema validity** across 15 videos, verified independently by three members. Gold set half-labelled. Repo, CI and a deployed Cloud Run "hello" endpoint live. **Go/no-go declared 22 Sep.**
2 23–29 Sep	End-to-end and ugly.	A command-line run takes a task plus photo and emits a merged procedure JSON. No UI. Alignment may be mediocre. **Nothing is demoed to anyone this week.**
3 30 Sep–6 Oct	Make it good, then stop adding.	Alignment hits target. Conflict typing works on at least four of six types. Frontend renders a real procedure with citations. Deployed and reachable on a public URL. **Scope freezes 6 Oct.**
4 7–13 Oct	Prove it, verify it, harden it.	Full eval run against all three baselines, numbers in BigQuery. Latency and cost tables populated from real traces. All 15 tasks cached. Deck drafted. Throttling live. **Verification loop shipped:** staged photo set captured, endpoint live, false-pass rate measured. If the verifier is not trustworthy by 13 Oct, it ships hidden behind a flag rather than half-working in the demo.
5 14–18 Oct	Ship the artifacts.	Three-minute video recorded and uploaded. README complete with architecture diagram and reproduction steps. Repo public. Deck as PDF. **Submitted 16 Oct.**

11.2 Role assignment

Role	Owns	Notes
R1 Engine	Extraction and reconciliation prompts, schemas, conflict typology, the go/no-go spike	The highest-risk role. Should be whoever is strongest at getting structured output out of a model reliably.
R2 Platform	Cloud Run services, Pub/Sub fan-out, Firestore, caching, CI/CD, deployment, throttling, budget alerts	Owns the fact that the live link still works on 6 November. This is a real responsibility, not an afterthought.
R3 Interface	React app, progressive streaming, conflict badge design, timestamp player, verification capture flow, demo polish	Only 10% of the rubric, but it is what the judge *sees*, so it disproportionately shapes their reading of everything else.
R4 Proof & story	Gold set, eval harness, baseline runs, deck, three-minute video, README, submission logistics	**Not the junior job.** This role produces the evidence for the heaviest-weighted criterion and every artifact a reviewer actually opens. R1 and R4 pair on labelling.

THREE RULES WE AGREE TO NOW

1. **Scope freeze 6 October is absolute.** Good ideas after that date go into section 16. No exceptions, no "it's only a small change."
2. **Everyone commits code, but R4's artifacts are deliverables with the same status as the engine.** A brilliant prototype with a rushed video scores worse than a good prototype with a sharp one.
3. **Standups Monday and Thursday, 30 minutes, blockers only.** Any metric target missed two sprints running gets escalated to a scope cut rather than more effort.

12 Risk register

Risk	Likely	Impact	Own	Mitigation
Video extraction is not structurally reliable	Med	Fatal	R1	Attacked in sprint 1 with a hard go/no-go on 22 Sep. Fallback: transcript + keyframe extraction, narrower domain, smaller source count. Four weeks remain at the decision point.
Step alignment quality stalls	High	Major	R1	Two full weeks budgeted. Degradation path: fall back to exact-order matching with lower support-count confidence, and say so in the coverage report. Product still works, claims get weaker.
Cold latency makes the demo feel broken	High	Major	R2/R3	Progressive streaming from sprint 3. Pre-warmed cache for all gold-set tasks. Demo runs cached, with the cold number stated honestly on screen.
Video token spend exceeds budget	Med	Major	R2	Budget alerts before first commit. Aggressive fps sampling. Gemma triage removes ~70% of candidates. Public endpoint throttled per session.
YouTube API quota exhausted mid-sprint	Med	Moderate	R2	Cache every search from the first commit. Frozen local corpus for the gold set so eval never touches the API.
Model version changes mid-build	Med	Moderate	R1	Pin exact version strings in config. Do not chase a new release after 6 Oct regardless of how much better it looks.
Scope creep, particularly toward Candidate B features	High	Major	All	This document, section 2. The scope freeze. Section 16 as the release valve for good ideas.
Deployed link dies during evaluation (19 Oct – 6 Nov)	Med	Severe	R2	Min-instance ≥1 on the API service, uptime check with alerting, throttling, and a residual budget reserved specifically for the evaluation window. Checked twice weekly.
Eligibility problem — any member is a student, or under 21, or outside JAPAC	Low	Fatal	R4	**Verify all four members against the rules this week.** This edition is working professionals only, 21+, JAPAC-based, and eligibility can be checked even after shortlisting. Disqualification is retroactive.
Fresh-code eligibility challenged	Low	Fatal	R2	New repo initialised after the programme launch date. Clean commit history from an empty first commit. No importing of prior personal projects.
Verifier produces a false pass on a wrongly-done step	Med	Severe	R1	Tuned for ≤2% false-pass against a staged photo set, accepting a high abstention rate as the cost. Three-outcome design with `unsure` as a first-class verdict. If we cannot hit the target by 13 Oct it ships behind a flag and does not appear in the demo.
Scope temptation toward AR / mixed reality	Med	Major	All	Section 16 records the decision and the reasoning so it does not get relitigated in week 3. The short answer stays the same: Google ships it first-party, it is a rendering layer rather than Gen AI innovation, and a native AR app fights our Cloud Run / Firebase compliance story.
Safety failure — we emit a dangerous instruction	Low	Severe	R1	Deterministic safety pack in code, hard termination gates on gas, mains electrical and structural work, 100% safety-omission recall target, and a visible disclaimer on every procedure.

13 Submission checklist and eligibility rules

13.1 Required artifacts

Artifact	Own	Requirement and our standard
Working deployed link	R2	Live public URL, anonymous access, no sign-up wall. Must survive the evaluation window to 6 Nov. Pre-loaded with a cached example so a judge sees value in under ten seconds.
Video demo, under 3 minutes	R4	Public YouTube, Vimeo or Google Drive link. Hard limit — a 3:10 video risks being treated as non-compliant. See section 14.
Public GitHub repository	R2	Public at submission, not after. README with architecture diagram, setup steps, model versions pinned, and the eval queries. No secrets in history, verified with a scan.
Deck as PDF	R4	Problem, solution, architecture, **the metrics slide**, scale and impact. Lift figures 2 and 3 from this document directly.
Theme declaration	R4	Media, Content & Digital Experiences. One theme, one submission, no hedging.

13.2 Rules that can disqualify us

- **Fresh projects only.** Work started before the programme launch is ineligible. New repo, clean history.
- **One submission, one theme.** No parallel entries.
- **Working professionals only**, aged 21 or over, based in JAPAC. Students are excluded and a single student on the team disqualifies the whole team, including after shortlisting.
- **Team of 2–4, locked 11 October.** Only two members travel to Singapore if shortlisted, so decide who and why before the finalists announcement on 7 November rather than in the middle of it.
- **All materials in English**, including code comments and documentation.
- **Mandatory stack:** Gemini or Gemma models (or an agentic platform), deployed via Cloud Run or Firebase. We satisfy both, and the deck should say so explicitly on the architecture slide.

DO THIS IN THE NEXT 48 HOURS

1. Confirm all four members meet the eligibility rules in writing. This is the only risk in the register that cannot be mitigated later.
2. Register the team and reserve the name. Deadline 11 October, but do it now.
3. Create the GCP project, enable billing, **set the budget alert**, and initialise the empty public repo.
4. Start the extraction spike. Nothing else in sprint 1 matters until it returns a verdict.

14 Demo video beat sheet

Three minutes, hard limit. The structural mistake most teams make is spending ninety seconds on the problem. We spend twenty, because the problem is self-evident to anyone who has ever stood under a sink.

Time	Beat	Content and intent
0:00–0:16	The problem, shown	Screen recording of a real search: forty results. Cut to five thumbnails side by side. Voiceover opens **on the conflict, not the synthesis**: these five contradict each other, and you have no way to know which one matches your sink. **No talking heads, no title cards, no logo animation.**
0:16–0:36	The interaction	Type the task. Take the photo. The system states what it sees: PVC slip-joint, plastic nuts. User confirms. This is the moment a judge understands the product is different.
0:36–0:56	Working, visibly	Sources found, triaged, rejected with reasons, then read in parallel. Compressed but honest — a caption states a cold run takes about two minutes and this one is cached.
0:56–1:38	**Payoff one: the diff**	The procedure. Then two moments in sequence: **(a)** the red safety badge — the most-watched video skips isolating the supply, four others include it, we made it mandatory; **(b)** the situational branch — V1 says use tape, your photo shows a rubber washer where tape causes the leak, so we selected the other branch. One tap on a citation jumps to that exact second of the source.
1:38–2:02	**Payoff two: verification**	The new closing argument. Do a step **deliberately wrong on camera** — washer seated bevel-up. Photograph it. The system catches it and names the correction, citing that three sources show bevel-down. Then the line that matters: *nothing without a reconciled corpus can do this, because nothing else knows what correct looks like.*
2:02–2:14	Generality	Change the domain dropdown. Motorcycle, then agriculture. No code changed, three config files. Twelve seconds, no more.
2:14–2:44	Architecture and proof	Figure 3 for eight seconds, narrated in one sentence — tiered Gemma, Flash and Pro routing on Cloud Run and Firebase. Then the metrics slide: our numbers against the top-video, concatenation and prose-blend baselines, plus the verification false-pass rate. **This is the 40% criterion, on screen, as numbers.**
2:44–2:58	Scale	One sentence on the domains this generalises to and the enterprise wedge. End on the positioning line: **we tell you where your sources contradict each other, and which one is right for you.**

PRODUCTION NOTES

- Record at 1440p or better. Judges may watch on a large screen and blurry UI reads as unfinished.
- Scripted voiceover recorded separately from screen capture. Live narration always overruns.
- Subtitles burnt in. Some evaluation happens with the sound off.
- Run a full rehearsal on 14 October. Recording on the 16th with no rehearsal is how teams end up submitting a 3:40 video.

15 Scale and commercial case

15.1 The generalisation claim

Crosscut applies wherever three conditions hold: procedural knowledge exists as unstructured video, multiple sources disagree, and the correct answer depends on the user's specific configuration. That is a large surface.

Domain	Situation dimension	Why the reconciliation matters
Home repair & maintenance v1 depth lane	Material, fitting, access	Wrong branch means water damage, a second failure and a paid call-out.
Vehicle & small engine	Model, year, market variant	Enormous informal video corpus, heavy model-year variation, high cost of error.
Agriculture strongest JAPAC impact case	Crop, soil, climate, pest	Guidance is highly situation-dependent and the best sources are frequently in the wrong language for the farmer who needs them.
Consumer electronics repair	Model, revision, part number	Board revisions change the procedure, and following the wrong revision destroys the device.
Field service & industrial maintenance the enterprise wedge	Equipment variant, site	Internal video libraries contradict official documentation. Reconciling them is a real operational problem people already pay to solve badly.
Clinical & allied health deliberately not v1	Protocol, jurisdiction	Highest value and highest validation burden. Correct to say we would only enter this with clinical partners.

15.2 Path beyond the prototype

- **Consumer** — free reconciliation with a paid tier for saved projects, offline procedures and multi-language output. Straightforward, low margin, useful as a distribution and data engine.
- **Enterprise** — the stronger business. Point the same engine at an organisation's internal training video library plus official documentation, and it produces one authoritative procedure per task while flagging every place the library contradicts the manual. That contradiction report is independently valuable to a training department and is a thing nobody currently has.
- **Platform** — reconciliation as an API for education platforms, marketplaces and equipment manufacturers who want verified procedures attached to the products they sell.

15.3 Why building this in JAPAC is the right framing

The competition asks for something built for a local need that scales globally. The situation-dependence problem is most acute precisely where the corpus and the user are mismatched: different hardware standards, different climate, different regulations, and a deep instructional corpus that exists mainly in English. A reconciliation layer is worth more to a farmer in Central Java or a technician in Cebu than to a homeowner in California, because the mismatch between the available content and their actual situation is larger. Build it where the problem is hardest and it works everywhere.

16 Out of scope for v1, and the v2 roadmap

This section exists so that good ideas have somewhere to go after the 6 October scope freeze. Adding something here is not a rejection.

Deferred item	Where it came from and when it returns
Voice interaction — hands-free step-through while actually working	Obvious and genuinely valuable: your hands are wet and under a sink. Chirp makes it straightforward. Deferred purely because it does not add rubric score for the effort. v2, high priority
Multi-language output — English corpus, local-language procedure	The single largest impact lever we are not pulling, and the strongest JAPAC story. One line in the deck as a roadmap item; not built. v2, high priority
True AR / mixed-reality overlay considered in v1.1 and deliberately rejected	Tempting, and wrong for this competition. Three reasons, in order of weight. **(1)** Live camera guidance with on-screen object highlighting is already a shipped first-party Google feature — demoing it as our innovation, to Google Cloud judges, is the weakest position available. **(2)** It is a rendering layer, so it earns little against "innovation in the use of Gen AI", which is where 25% of the score lives. **(3)** A native ARCore app fights the Cloud Run / Firebase deployment requirement, and WebXR will not run reliably on the laptop a judge clicks our link from. The verification loop captures the real value of "point your camera at it" at a fraction of the cost. rejected, not deferred
Physics or repair simulation	Also rejected. There is no plumbing physics worth simulating and a fake simulation is worse than none — a judge who spots it discounts everything else we claim. The honest version of the idea survives as consequence preview, which illustrates a *documented* failure mode from our own conflict data rather than pretending to model one. rejected, not deferred
Coverage gap-filling diagrams	The v1.0 use of Imagen. Traded away to fund verification, which is worth more per engineering hour. Imagen stays in the stack, repointed at consequence preview. v2
Generated video gap-filling with Veo	From the kids-education idea. Cost and latency put it out of reach for a four-week build. v2
Live continuous verification — streaming camera rather than a still photo	The natural extension of the verification loop and the place where a genuine camera-native experience becomes defensible, because by then we are verifying against a corpus rather than just describing a scene. Needs a live streaming model path and far more latency engineering than four weeks allows. v2, high priority
Provenance and attribution certificates	From the copyright brainstorm. Interesting applied to *procedures* rather than videos — a signed, versioned, cited procedure as a durable artifact. Closer to the enterprise product than the consumer one. v2/v3
Community correction loop	Users who complete a procedure report what was wrong, which updates the conflict resolution. Real moat, meaningless without users. post-hackathon
Parental media-diet surface (Candidate B)	Same engine, different declared constraint. The most natural second product and the reason the architecture is worth building generally. v2, separate surface
Clinical and allied health domain	Requires clinical validation partners. Deliberately excluded, and saying so out loud reads as judgement rather than limitation. not without partners

A Appendix — StepGraph JSON schema

The contract every extraction call must satisfy. Schema-constrained generation, validated before anything downstream sees it. Invalid output is retried once, then the source is dropped and recorded as a coverage gap.

```
{
  "source": {
    "video_id": "string",
    "url": "string",
    "duration_s": 0,
    "language": "string",
    "channel": "string",
    "view_count": 0
  },
  "domain": "home_repair",
  "task_label": "replace P-trap under kitchen sink",
  "is_procedural": true,

  "situation_markers": {
    "material":   "pvc | abs | copper | pex | galvanised | unknown",
    "joint_type": "slip_joint | compression | threaded | soldered | push_fit | unknown",
    "access":     "open_cabinet | wall_cavity | crawl_space | unknown",
    "confidence": 0.0
  },

  "requirements": {
    "tools":     [{"name": "string", "essential": true, "seen_at_s": 0.0}],
    "materials": [{"name": "string", "quantity": "string", "spec": "string"}]
  },

  "steps": [
    {
      "id": "s1",
      "order": 1,
      "action": "verb phrase, imperative",
      "object": "what is acted on",
      "detail": "one sentence, model's own words",
      "preconditions": ["free-text, aligns to earlier step ids where possible"],
      "t_start_s": 0.0,
      "t_end_s": 0.0,
      "evidence_modality": "visual | spoken | on_screen_text | multiple",
      "evidence_note": "e.g. 'hand rotating oval valve handle clockwise'",
      "applies_when": {"material": ["pvc"], "joint_type": ["slip_joint"]},
      "safety": {
        "level": "none | caution | critical | terminate",
        "note": "string"
      },
      "expected_post_state": {
        "description": "what this looks like when done correctly, as seen from outside",
        "visual_checks": ["bevel of the washer faces down",
                          "nut hand-tight, threads not visible"],
        "common_errors":  [{"error": "washer seated bevel-up",
                            "visible_as": "raised lip visible above the nut",
                            "consequence": "slow drip under load"}],
        "verifiable_from_photo": true
      },
      "confidence": 0.0
    }
  ],

  "suspected_omissions": [
    {"expected_step": "string", "reason": "string"}
  ],

  "quality_signals": {
    "hands_on_camera": true,
    "sponsored_spans_s": [[0.0, 0.0]],
    "audio_only_spans_s": [[0.0, 0.0]],
    "overall_instructional_quality": 0.0
  }
}
```

Design notes on the schema

- `applies_when` **is the personalisation hinge.** If extraction populates it well, situation-dependent conflicts resolve almost automatically. If it comes back mostly empty, the aligner has to infer applicability, which is much harder. Watch this field closely during the sprint-1 spike.
- `evidence_note` **exists to catch hallucination.** A model that must describe what it saw at a timestamp is measurably less likely to invent the step. It also gives us a cheap citation-validity audit.
- `suspected_omissions` **is speculative by design.** A single source claiming another step should exist is weak evidence; the same claim from three sources is exactly how safety-critical omissions get caught.
- `sponsored_spans_s` lets us exclude advertising segments from step extraction, which is a real and otherwise noisy failure mode.

- `expected_post_state` **is what makes verification possible**, and it is the v1.1 addition to this schema. Note that `common_errors` is populated from what sources *warn about*, not from the model's imagination — when three videos all caution against seating a washer upside down, that is exactly the check worth running against the user's photo. `verifiable_from_photo` lets the UI hide the verify button on steps where a still image could not settle the question.

B Appendix — prompt skeletons

Starting points only. Expect these to be rewritten many times; keep them versioned in the repo with the eval score attached to each version so we can prove improvement rather than assert it.

B.1 Extractor

```
ROLE
You are a procedural analyst. You convert instructional video into a
structured step graph. You do not summarise and you do not advise.

INPUT
A video, plus the domain situation taxonomy.

RULES
1  Emit ONLY steps that are demonstrated or explicitly stated in this video.
   Never add a step because it is good practice. Omissions are data.
2  Every step needs a timestamp locating where it is demonstrated, and an
   evidence_note describing what is visible or said at that moment.
3  Populate applies_when only from what this video shows about the hardware.
   Leave a dimension out rather than guessing it.
4  Mark safety level from what the video does, not from your own judgement of
   risk. If the video never mentions isolating the supply, do not invent it.
5  If the video is not procedural, set is_procedural false and stop.
6  Output valid JSON conforming to the StepGraph schema. Nothing else.
```

B.2 Reconciler

```
ROLE
You reconcile several independently extracted step graphs for one task into a
single procedure, and you classify every disagreement between them.

INPUT
N StepGraph objects. One SituationProfile for this user. The domain safety pack.

RULES
1  Work from the provided step clusters. Do not re-align steps yourself.
2  For every cluster where sources differ, assign exactly ONE type:
     EQUIVALENT_ALTERNATIVE | SITUATION_DEPENDENT | GENUINE_DISAGREEMENT
     OUTDATED_PRACTICE      | ORDERING_VARIANT    | SAFETY_CRITICAL_OMISSION
3  Justify each type with evidence quoted from BOTH sides, citing video_id and
   timestamp. A conflict you cannot evidence is not a conflict; drop it.
4  SITUATION_DEPENDENT: state which branch the SituationProfile selects and why.
   If the profile does not adjudicate, downgrade to GENUINE_DISAGREEMENT rather
   than choosing for the user.
5  SAFETY_CRITICAL_OMISSION: flag it. Do NOT decide what happens next --
   the safety pack is applied deterministically downstream.
6  Never invent a step absent from all sources. If the procedure has a hole,
   record it in the coverage report and leave it open.
7  Order steps so no precondition is violated. Where order is free, choose the
   safest sequence.
```

B.3 Verifier

```
ROLE
You inspect one photograph of work a user has just completed and judge it
against an expected post-state derived from multiple independent sources.

INPUT
One photo. The step's expected_post_state (description, visual_checks,
common_errors). The situation profile.

RULES
1  Return exactly one verdict: PASS | FAIL | UNSURE.
2  UNSURE is not a failure of yours -- it is the correct answer whenever the
   photo is too dark, too far, occluded, or taken from an angle that cannot
   settle the question. Say what to change: "retake from the left, closer."
3  Never PASS on the basis of what is probably true. A false PASS sends the
   user forward confident and wrong; an unnecessary UNSURE costs them ten
   seconds. The asymmetry is enormous, so bias hard toward UNSURE.
4  On FAIL, name the specific error from common_errors where one matches, and
   describe what you can actually see that indicates it. Cite the sources that
   warn about it.
5  Judge only this step's post-state. Do not comment on unrelated things
   visible in the photo, and do not offer to help with later steps.
6  Never infer from the photo that the overall task is finished or safe.
```

WHY RULE 5 IN THE RECONCILER MATTERS

The model identifies that a safety step was omitted; **code** decides whether that becomes a warning or a hard gate. Keeping that boundary crisp is both good engineering and a strong answer when a judge asks how we handle model error in a safety-relevant product.

C Appendix — repository structure

```
crosscut/
  README.md                 # architecture diagram, setup, pinned model versions,
                            # eval reproduction steps. Judges read this first.
  .github/workflows/        # test, build, deploy on merge to main
  infra/
    cloudbuild.yaml
    terraform/              # optional; Cloud Run + Firestore + Pub/Sub
  domains/
    home_repair.yaml        # depth lane
    motorcycle.yaml         # breadth lane
    agriculture.yaml        # breadth lane
  services/
    api/                    # Cloud Run: FastAPI, SSE, agent graph entrypoint
      agents/
        profiler.py
        retriever.py
        triage.py
        extractor.py
        aligner.py
        reconciler.py
        personaliser.py
        gapfiller.py
      schemas/              # pydantic models mirroring appendix A
      safety/               # deterministic gates -- NOT prompts
      prompts/              # versioned, each tagged with its eval score
    worker/                 # Cloud Run Jobs: extraction fan-out consumer
    triage-model/           # Gemma container
  web/                      # React + Vite + Tailwind, Firebase Hosting
  eval/
    goldset/                # 15 tasks, frozen video ids, reference step lists
    baselines/              # top-video, concatenation, prose-blend
    run_eval.py
    queries/                # the BigQuery SQL behind every deck number
  docs/
    proposal.pdf            # this document
    architecture.svg
```

README STANDARD

Assume a judge spends ninety seconds in the repo. The README must, in that time, show the architecture diagram, state which Google models are used and where, link the live deployment, and show the metrics table with a command to reproduce it. Everything else is secondary.

D Appendix — glossary

Term	Meaning in this document
Step graph	The machine-readable representation of one video's procedure: ordered steps with tools, preconditions, timestamps, applicability and safety level. Appendix A.
Situation profile	The structured model of the user's specific case, extracted from a photo plus at most two clarifying questions. What makes personalisation possible.
Alignment	Recognising that steps phrased differently across sources are the same physical action. Prerequisite for counting agreement.
Step cluster	The set of steps from different sources that alignment has determined are the same step. Its size is the support count.
Support count	How many independent sources contain a given step. Drives confidence display and omission detection.
Conflict type	One of six categories assigned to every divergence. Section 4.3. Our central original contribution.
Safety pack	Per-domain deterministic rules that decide what happens when a safety-relevant condition is detected. Code, never a prompt.
Expected post-state	What a correctly-completed step looks like from outside, with named visual checks and the errors sources warn about. Extracted during step-graph extraction; the input to verification. Appendix A.
Verification loop	The user photographs completed work; the system judges it against the expected post-state and returns pass, a named correction, or an honest abstention. Only possible given a reconciled corpus, which is why it is our most defensible feature.
Consequence preview	A generated illustration of what goes wrong on the branch we rejected, driven by conflict data. The only honest reading of "simulation" in this product: we illustrate a documented failure mode rather than modelling physics.
Coverage report	The honest account of which steps rest on thin evidence and which parts of the task no source covered. Honesty as a product feature.
Depth lane / breadth lane	The one domain we tune and measure, versus the domains we prove generality with via config swap. Section 5.
Gemini Enterprise Agent Platform	Google Cloud's AI platform, renamed from Vertex AI in April 2026. Use the current name in judge-facing material.

DOCUMENT STATUS

Version 1.1, 16 September 2026. Sections 1, 3, 4, 5 and 15 plus figures 2 and 3 are the judge-facing material. Sections 6 to 12 and the appendices are the engineering specification.

v1.1 changes: verification loop added (4.1, 4.6, 6, 7, 9, 11, 14, appendices A and B); Imagen repointed from coverage gap-filling to consequence preview; competitive position against live camera assistants added as 4.5; AR/MR and physics simulation explicitly rejected with reasoning in section 16; positioning sharpened to lead with the conflict rather than the synthesis.

Open decisions: final product name (by 18 Sept), and confirmation of the go/no-go spike result (by 22 Sept).